DeepWAF-XAI: Statistical Comparison Tables

Summary of individual literature baselines, calculated arithmetic means (=AVERAGE formulas), and m

Table 2.1: Accuracy Benchmark Comparison

System / Baseline Architecture	enchmark Source / Literature Referenc
Traditional Rule-Based WAF (ModSecurity CRS)	OWASP ModSecurity Core Rule Set Benchmar
Classical Machine Learning Baseline (SVM / RF)	Standard CSIC 2010 N-Gram TF-IDF Classifier
BALNet (CNN Base Paper, Springer 2026)	Springer Discover Applied Sciences (2026) 8:2
UniEmbed (Feature Fusion MLP, Springer 2025)	Springer (2025) Payload Feature Fusion Bench
HawkPhish-DNN (HHO-DNN, Springer 2025)	Springer Cluster Computing (2025) Phishing B
Existing Systems (Arithmetic Mean)	**Excel Calculated Arithmetic Mean of Availa**
Proposed DeepWAF-XAI	**Empirical Measurement on Held-out Test S**

Table 2.2: Precision Benchmark Comparison

System / Baseline Architecture	enchmark Source / Literature Referenc
Traditional Rule-Based WAF (ModSecurity CRS)	OWASP ModSecurity Core Rule Set Benchmar
Classical Machine Learning Baseline (SVM / RF)	Standard CSIC 2010 N-Gram TF-IDF Classifier
BALNet (CNN Base Paper, Springer 2026)	Springer Discover Applied Sciences (2026) 8:2
UniEmbed (Feature Fusion MLP, Springer 2025)	Springer (2025) Payload Feature Fusion Bench
HawkPhish-DNN (HHO-DNN, Springer 2025)	Springer Cluster Computing (2025) Phishing B
Existing Systems (Arithmetic Mean)	**Excel Calculated Arithmetic Mean of Availa**
Proposed DeepWAF-XAI	**Empirical Measurement on Held-out Test S**

Table 2.3: Recall Benchmark Comparison

System / Baseline Architecture	enchmark Source / Literature Referenc
Traditional Rule-Based WAF (ModSecurity CRS)	OWASP ModSecurity Core Rule Set Benchmar
Classical Machine Learning Baseline (SVM / RF)	Standard CSIC 2010 N-Gram TF-IDF Classifier
BALNet (CNN Base Paper, Springer 2026)	Springer Discover Applied Sciences (2026) 8:2
UniEmbed (Feature Fusion MLP, Springer 2025)	Springer (2025) Payload Feature Fusion Bench
HawkPhish-DNN (HHO-DNN, Springer 2025)	Springer Cluster Computing (2025) Phishing B
Existing Systems (Arithmetic Mean)	**Excel Calculated Arithmetic Mean of Availa**
Proposed DeepWAF-XAI	**Empirical Measurement on Held-out Test S**

Table 2.4: F1-Score Benchmark Comparison

System / Baseline Architecture	enchmark Source / Literature Referenc
Traditional Rule-Based WAF (ModSecurity CRS)	OWASP ModSecurity Core Rule Set Benchmar
Classical Machine Learning Baseline (SVM / RF)	Standard CSIC 2010 N-Gram TF-IDF Classifier
BALNet (CNN Base Paper, Springer 2026)	Springer Discover Applied Sciences (2026) 8:2
UniEmbed (Feature Fusion MLP, Springer 2025)	Springer (2025) Payload Feature Fusion Bench

HawkPhish-DNN (HHO-DNN, Springer 2025)	Springer Cluster Computing (2025) Phishing B
Existing Systems (Arithmetic Mean)	**Excel Calculated Arithmetic Mean of Availa**
Proposed DeepWAF-XAI	**Empirical Measurement on Held-out Test S**

Table 2.5: Category-Level Evaluation Comparison Across All Metrics (Table 8.4 in

Attack Category / Evaluation Scope	Metric
Normal Traffic (LEGAL)	Accuracy
Normal Traffic (LEGAL)	Precision
Normal Traffic (LEGAL)	Recall
Normal Traffic (LEGAL)	F1-Score
SQL Injection (SQL)	Accuracy
SQL Injection (SQL)	Precision
SQL Injection (SQL)	Recall
SQL Injection (SQL)	F1-Score
Cross-Site Scripting (XSS)	Accuracy
Cross-Site Scripting (XSS)	Precision
Cross-Site Scripting (XSS)	Recall
Cross-Site Scripting (XSS)	F1-Score
Path Traversal (SHELL)	Accuracy
Path Traversal (SHELL)	Precision
Path Traversal (SHELL)	Recall
Path Traversal (SHELL)	F1-Score
Phishing URLs (PHISHING)	Accuracy
Phishing URLs (PHISHING)	Precision

Phishing URLs (PHISHING)	Recall
Phishing URLs (PHISHING)	F1-Score
Overall Average	**Accuracy**
Overall Average	**Precision**
Overall Average	**Recall**
Overall Average	**F1-Score**

easured DeepWAF-XAI results.

Accuracy (%)	ion Status / Comparability
82.40	Published literature baseline (Regex pattern matching)
88.70	Published literature baseline (Tabular statistical ML)
91.30	Published literature baseline (1D CNN with ADASYN)
99.82	Published literature baseline (Binary payload classification)
99.60	Published literature baseline (URL phishing classification)
92.36	**Aggregated Existing Systems Baseline Line Value**
92.59	**Project-Measured Result (Overall Support-Weighted)**

Precision (%)	ion Status / Comparability
84.10	Published literature baseline (Regex pattern matching)
89.20	Published literature baseline (Tabular statistical ML)
92.00	Published literature baseline (1D CNN with ADASYN)
99.80	Published literature baseline (Binary payload classification)
99.80	Published literature baseline (URL phishing classification)
92.98	**Aggregated Existing Systems Baseline Line Value**
97.14	**Project-Measured Result (Overall Support-Weighted); Macro: 58.23%**

Recall (%)	ion Status / Comparability
79.60	Published literature baseline (Regex pattern matching)
86.50	Published literature baseline (Tabular statistical ML)
90.80	Published literature baseline (1D CNN with ADASYN)
99.86	Published literature baseline (Binary payload classification)
99.40	Published literature baseline (URL phishing classification)
91.23	**Aggregated Existing Systems Baseline Line Value**
92.59	**Project-Measured Result (Overall Support-Weighted); Macro: 89.43%**

F1-Score (%)	ion Status / Comparability
81.80	Published literature baseline (Regex pattern matching)
87.80	Published literature baseline (Tabular statistical ML)
91.40	Published literature baseline (1D CNN with ADASYN)
99.83	Published literature baseline (Binary payload classification)

99.60	Published literature baseline (URL phishing classification)
92.09	**Aggregated Existing Systems Baseline Line Value**
94.32	**Project-Measured Result (Overall Support-Weighted); Macro: 65.93%**

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ng Systems (Averag	roposed DeepWAF-XAI (%	Benchmark Basis / Academic Notes
91.10	92.40	ModSecurity CRS (82.4%) & BALNet Benign (99.79%). DeepWAF-XAI achieves 92.40% accuracy.
92.05	99.92	DeepWAF-XAI delivers 99.92% precision, effectively eliminating false positive blocking.
92.50	92.40	Directly comparable legitimate request passage rate.
92.25	96.01	DeepWAF-XAI harmonic score exceeds existing systems average by +3.76%.
90.56	77.58	Evaluated on severe natural minority class (165 SQL test samples vs 18,290 Normal).
89.48	14.55	Natural imbalance causes low raw precision; mitigated by secondary DB firewall.
90.24	77.58	DeepWAF-XAI intercepts 128 / 165 SQL injection attacks on first perimeter pass.
89.84	24.50	Harmonic score under natural production class skew.
90.56	95.94	DeepWAF-XAI exceeds existing systems average by +5.38%.
80.78	56.24	Exceeds published BALNet XSS precision (50.0%).
74.74	95.94	DeepWAF-XAI intercepts 757 / 789 XSS scripts (+21.20% higher recall than existing avg).
77.36	70.91	High sequence sensitivity to script injection tags and obfuscated events.
85.55	81.25	ModSecurity CRS & ML baseline. DL papers (BALNet, UniEmbed) report N/A for Path Traversal.
86.65	28.89	Extreme minority class (32 samples out of 19,776).
83.05	81.25	DeepWAF-XAI intercepts 26 / 32 path traversal and directory traversal attempts.
84.80	42.62	Represents robust detection under severe sample sparsity.
99.60	100.00	DeepWAF-XAI intercepts 500 / 500 test phishing URLs (100.00% accuracy).
99.80	91.58	High precision against deceptive domain spoofing.

99.40	100.00	Perfect recall boundary on held-out test split, exceeding HawkPhish-DNN.
99.60	95.60	Strong harmonic detection balance.
92.36	**92.59**	**DeepWAF-XAI achieves 92.59% overall accuracy (18,311 / 19,776 correct predictions).**
92.98	**97.14**	**DeepWAF-XAI Support-Weighted Precision is 97.14% (Macro: 58.23%).**
91.23	**92.59**	**DeepWAF-XAI Support-Weighted Recall is 92.59% (Macro: 89.43%).**
92.09	**94.32**	**DeepWAF-XAI Support-Weighted F1-Score is 94.32% (Macro: 65.93%).**